

CHARLES DARWIN UNIVERSITY

PRT661 · Data Science Practice

Early vs Late Potato Blight Classification

Group: Dan7– Theme2

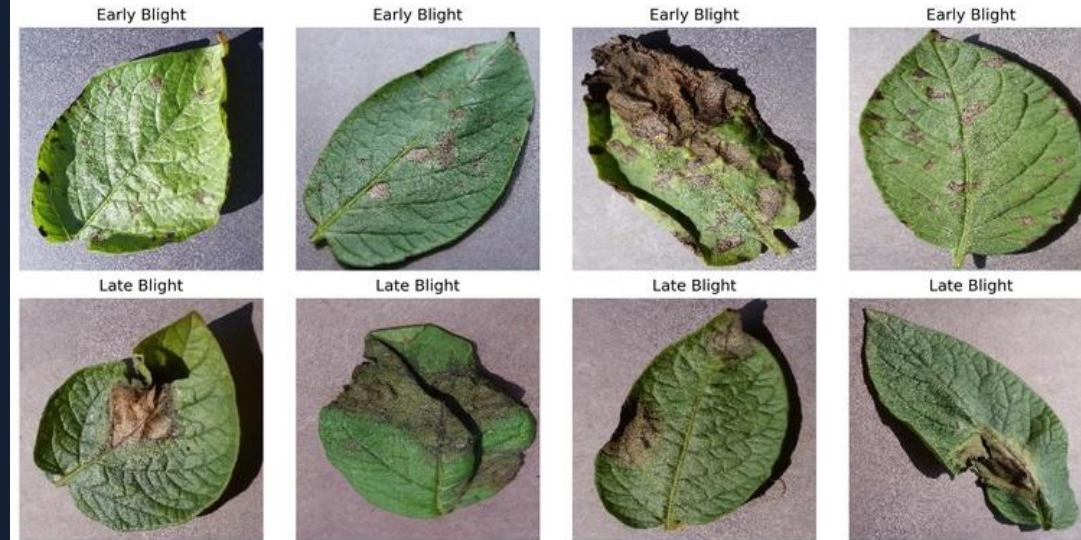
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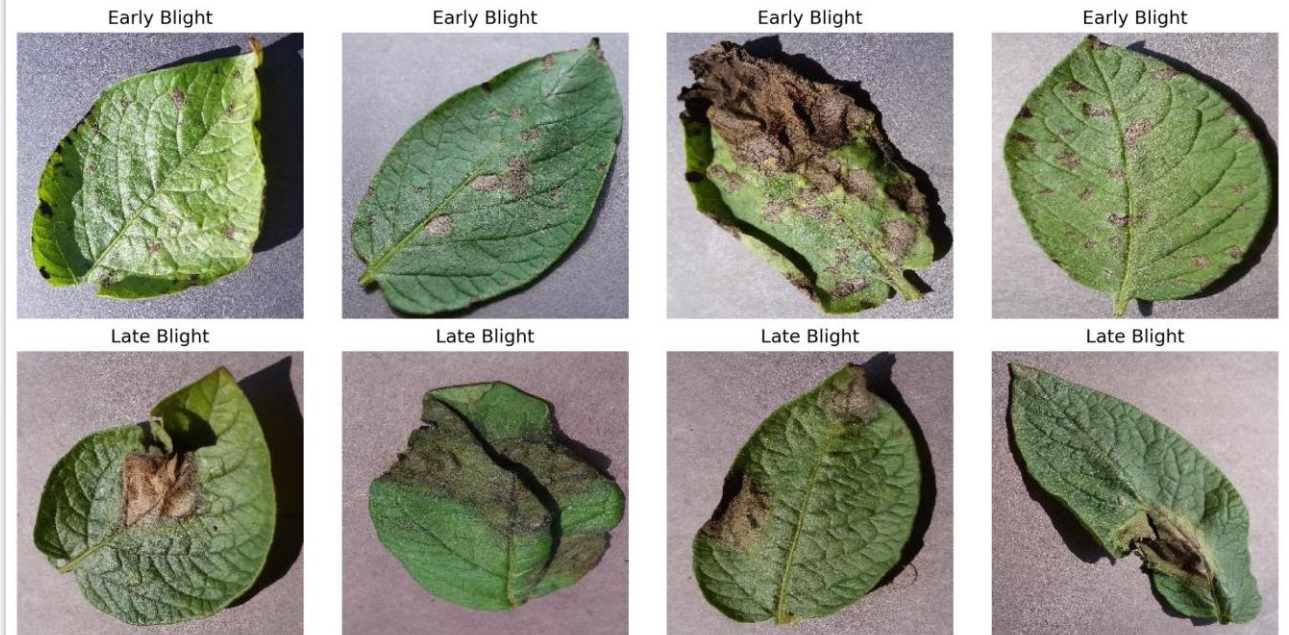
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Problem and research question

- Early and Late Blight can produce visually confusing leaf symptoms.
- Random image splitting can overstate performance when multiple images come from the same leaf.
- Question: can leakage-controlled models separate the two labels and retain useful performance on untouched PlantDoc images?
- Scope: Early vs Late Blight only; no Healthy, Other, pesticide or treatment output.



Balanced dataset and fixed split

TOTAL

2,000

RGB images

EARLY BLIGHT

1,000

balanced class

LATE BLIGHT

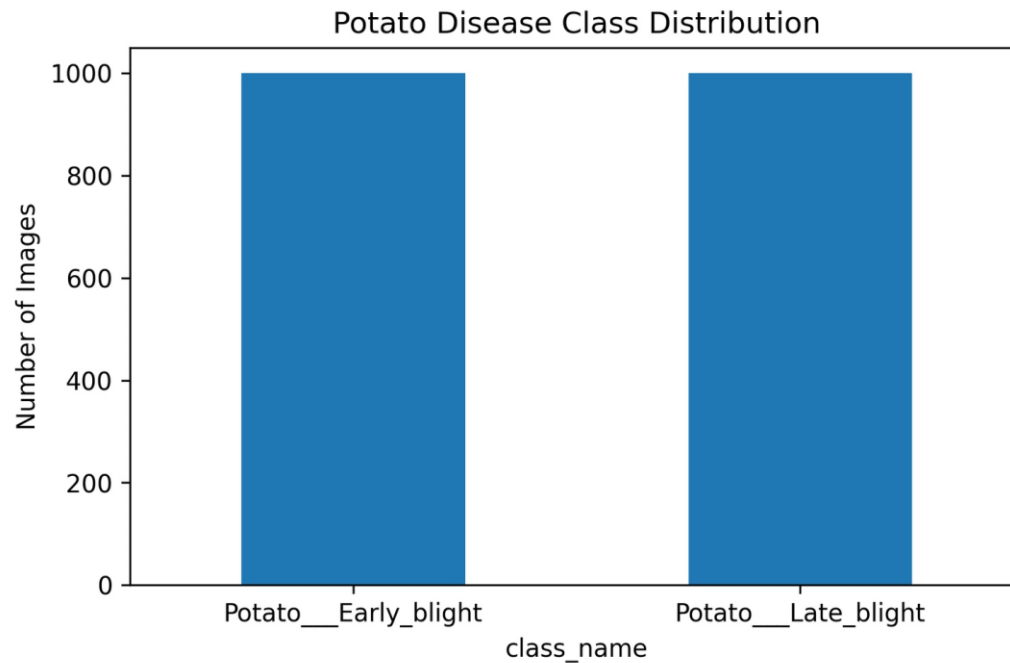
1,000

balanced class

LEAF GROUPS

500

group-aware audit



Train / validation / test = 1,400 / 300 / 300 · 350 / 75 / 75 disjoint leaf groups · zero shared groups

Implemented workflow

1

Audit

Folders, classes, dimensions and files

2

Split

Group-disjoint 350 / 75 / 75 leaf groups

3

Transform

224×224; training-only augmentation

4

Train

Logistic + PotatoCNN + MobileNetV2

5

Select

Validation-only checkpoints + thresholds

6

Test

Locked PlantVillage + untouched
PlantDoc tests

7

Diagnose

Errors, XAI, calibration, robustness and
McNemar

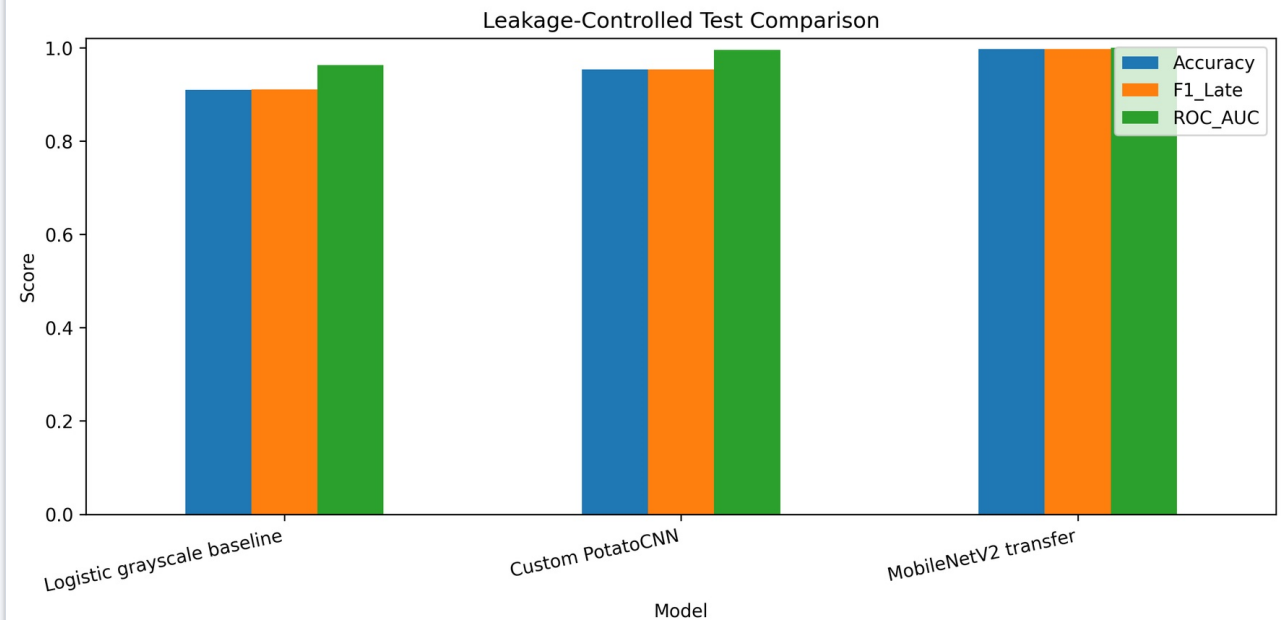
8

Communicate

Versioned evidence interface and report

Three-model design

- Logistic grayscale baseline · traditional reference · threshold 0.70.
- PotatoCNN · 93,377 parameters · 224×224 RGB · threshold 0.38.
- MobileNetV2 · ImageNet transfer learning with staged fine-tuning · threshold 0.64.
- All operating thresholds were selected from validation data before test evaluation.



MobileNetV2 training and checkpoint selection

EPOCHS

8

head + fine-tune

BEST VAL F1

0.9967

epoch 8

VAL LOSS

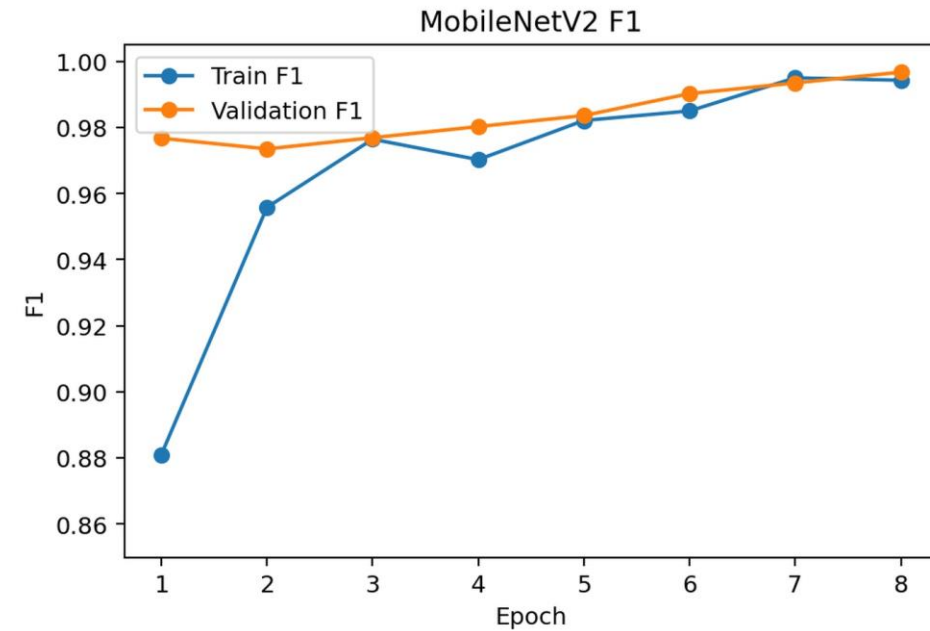
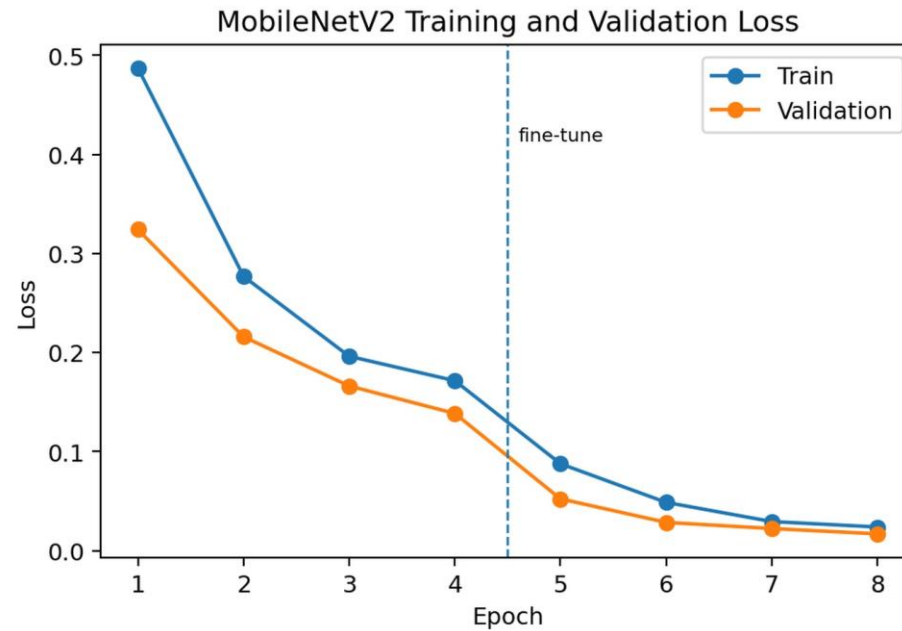
0.0169

final fine-tune stage

LOCKED THRESHOLD

0.64

validation-only selection



Primary MobileNetV2 held-out result

ACCURACY

99.67%

299 / 300

PRECISION

100.00%

Late-Blight predictions

RECALL

99.32%

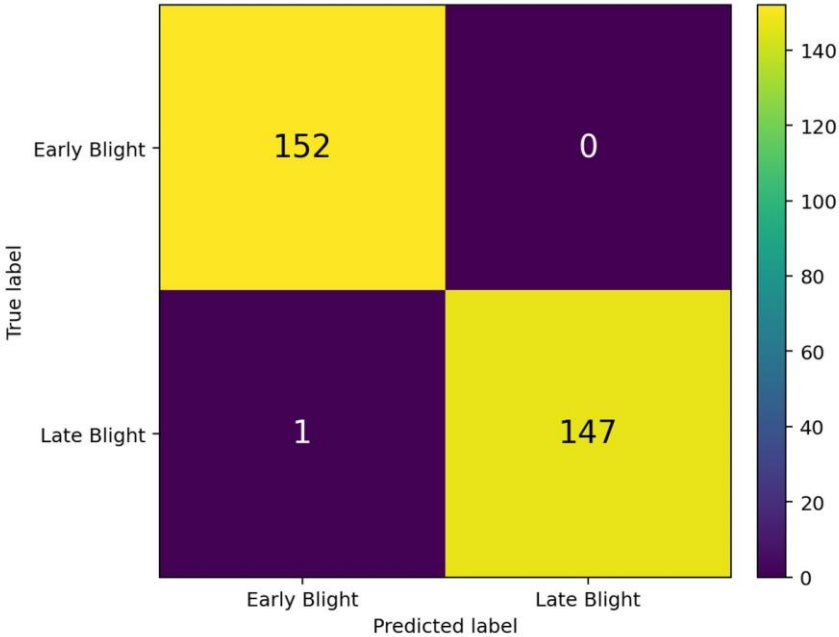
147 / 148 Late

ROC-AUC

1.0000

group-isolated internal set

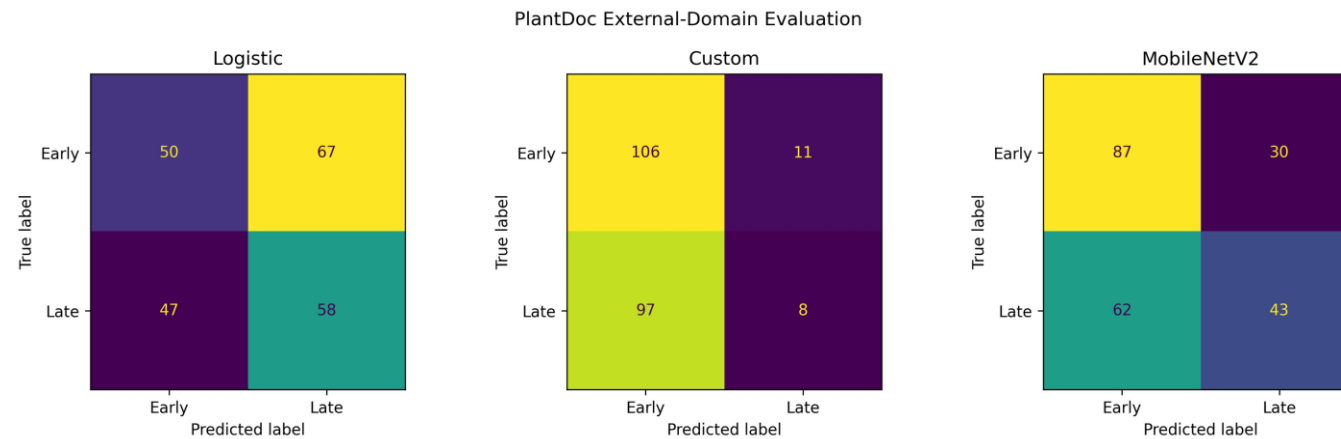
MobileNetV2 Internal Test Confusion Matrix



One error: 1 Late→Early · 0 Early→Late

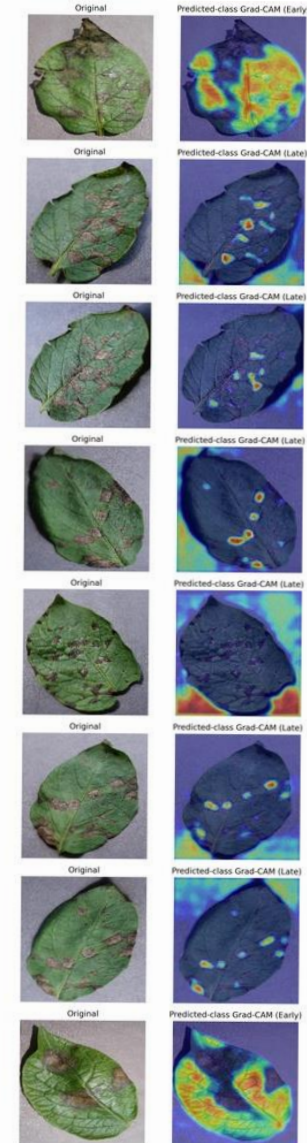
Untouched PlantDoc external evaluation

- 222 external PlantDoc images were evaluated with no fitting, retuning or threshold selection.
- MobileNetV2: accuracy 0.5856 · Late F1 0.4831 · ROC–AUC 0.6210.
- PotatoCNN: accuracy 0.5135 · Late F1 0.1290; logistic accuracy 0.4865.
- The large internal-to-external drop defines the main domain-generalisation limitation.



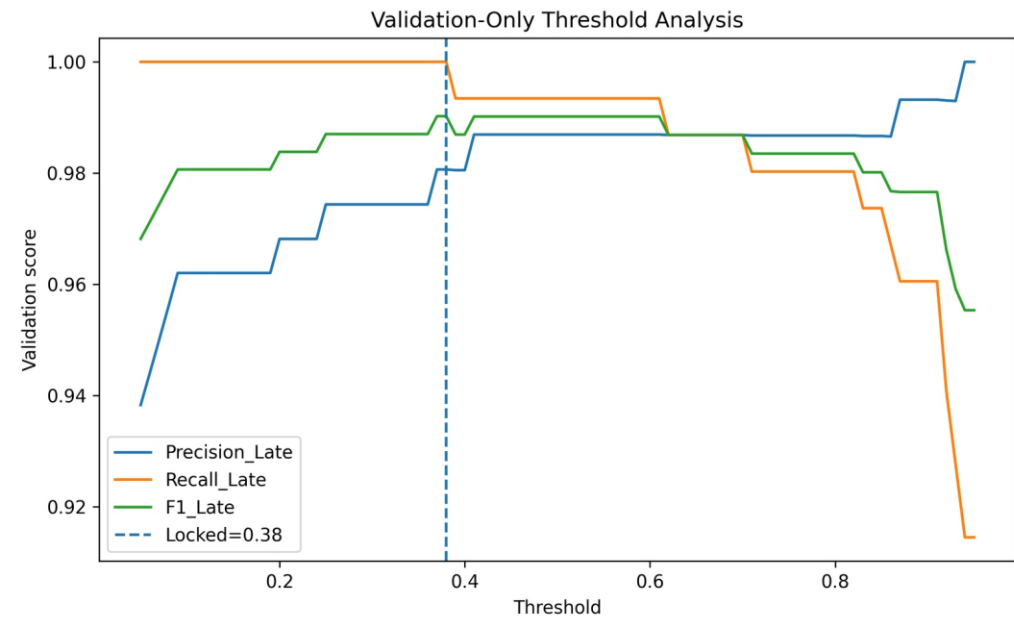
Explainability and internal error review

- Grad-CAM highlights regions influencing the class score in representative cases.
- Attention often overlaps lesions but can also include edges, background and broad leaf structure.
- PCA/t-SNE provide exploratory feature-separation views inside the controlled source.
- Heatmaps are diagnostic evidence—not pathogen confirmation or causal lesion localisation.



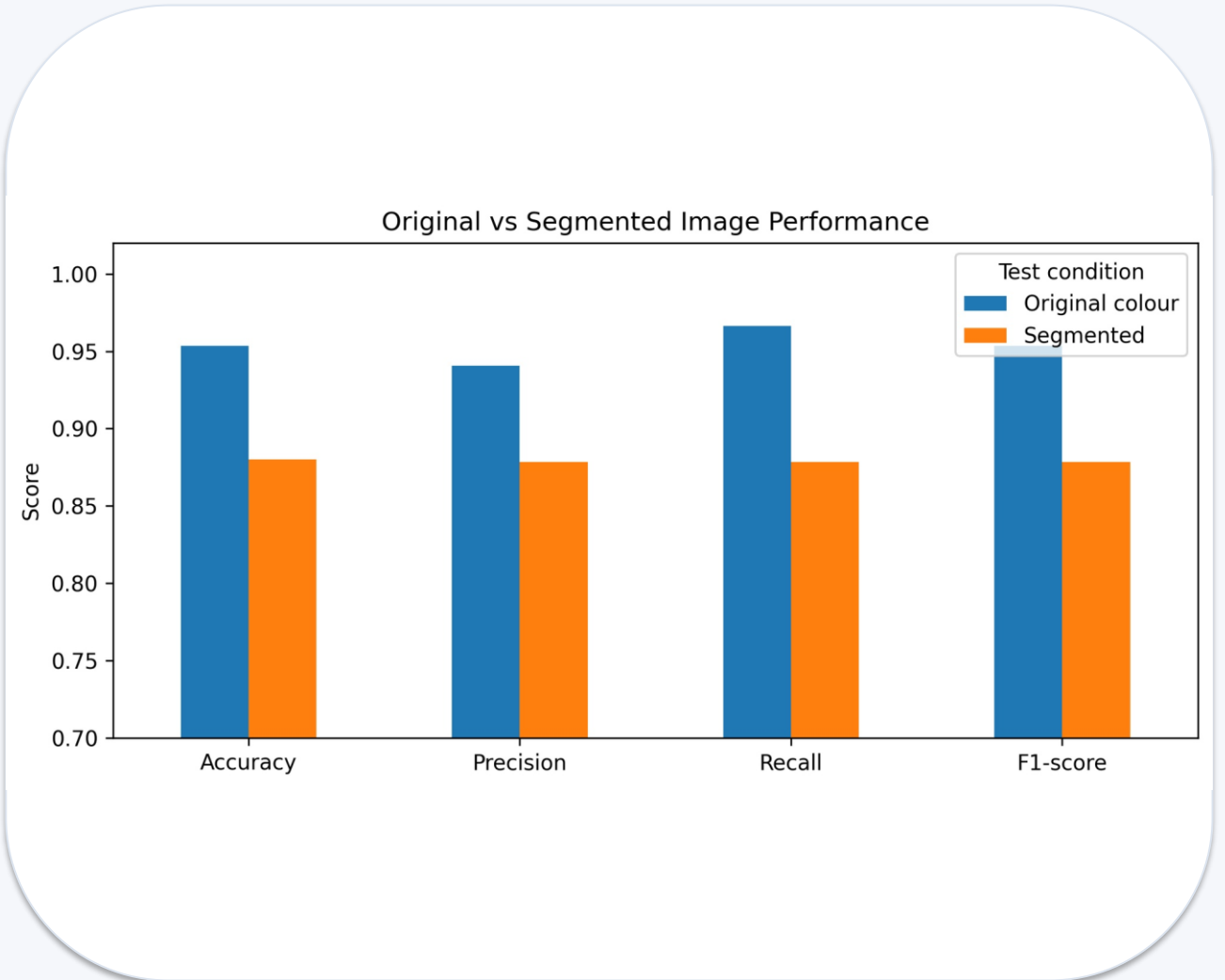
Validation thresholds and paired evidence

- Locked thresholds: logistic 0.70 · PotatoCNN 0.38 · MobileNetV2 0.64.
- PotatoCNN threshold 0.38 was selected from validation data only; no PlantDoc tuning.
- McNemar: Logistic vs MobileNetV2 $p=2.16 \times 10^{-7}$; PotatoCNN vs MobileNetV2 $p=0.00098$.
- Threshold curves are diagnostic evidence, not permission for test-set optimisation.



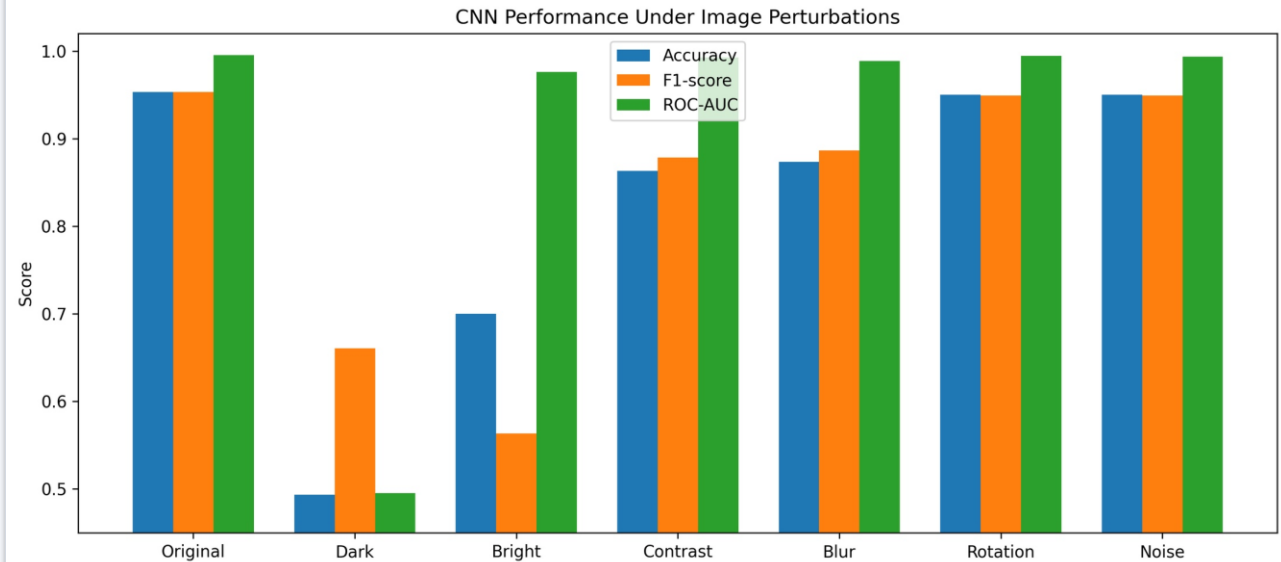
Segmentation exposed a probability shift

- At the ordinary 0.50 rule, segmented images produced no Late-Blight decisions.
- Segmented ROC–AUC remained 0.9566 despite the probability-scale shift.
- A threshold selected only on segmented validation recovered accuracy 0.8800 and F1 0.8784.
- Preprocessing + model + threshold must be validated and versioned as one system.



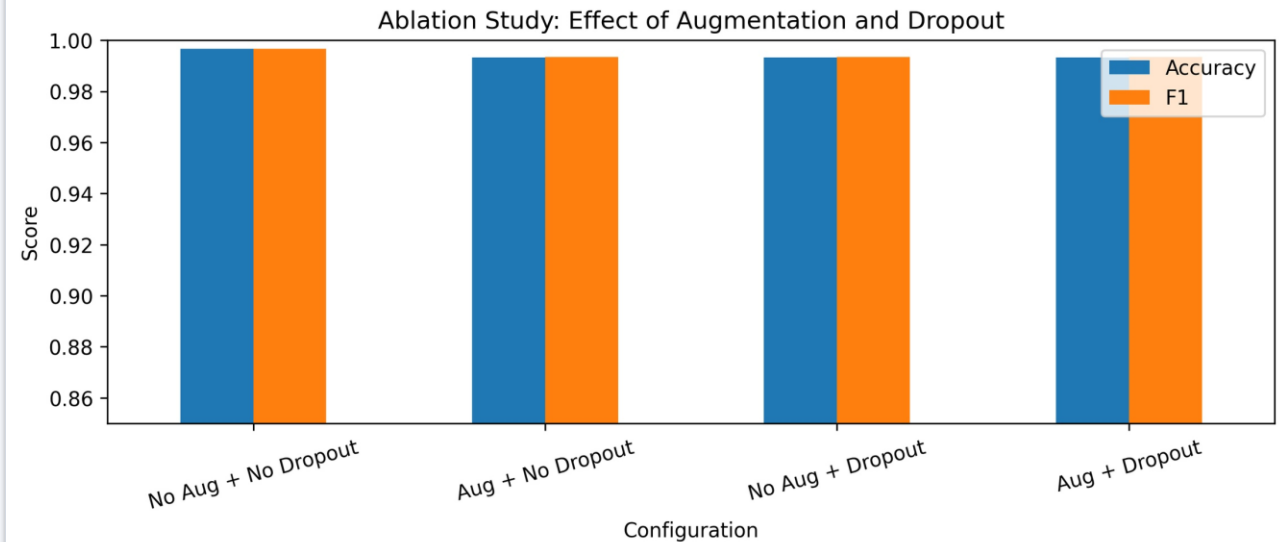
Controlled image perturbations

- Dark F1 0.6607 · Bright F1 0.5631.
- Contrast 0.8783 · Blur 0.8862.
- Rotation 0.9492 · Noise 0.9492.
- Strong ROC–AUC can coexist with poor fixed-threshold performance after distribution shift.

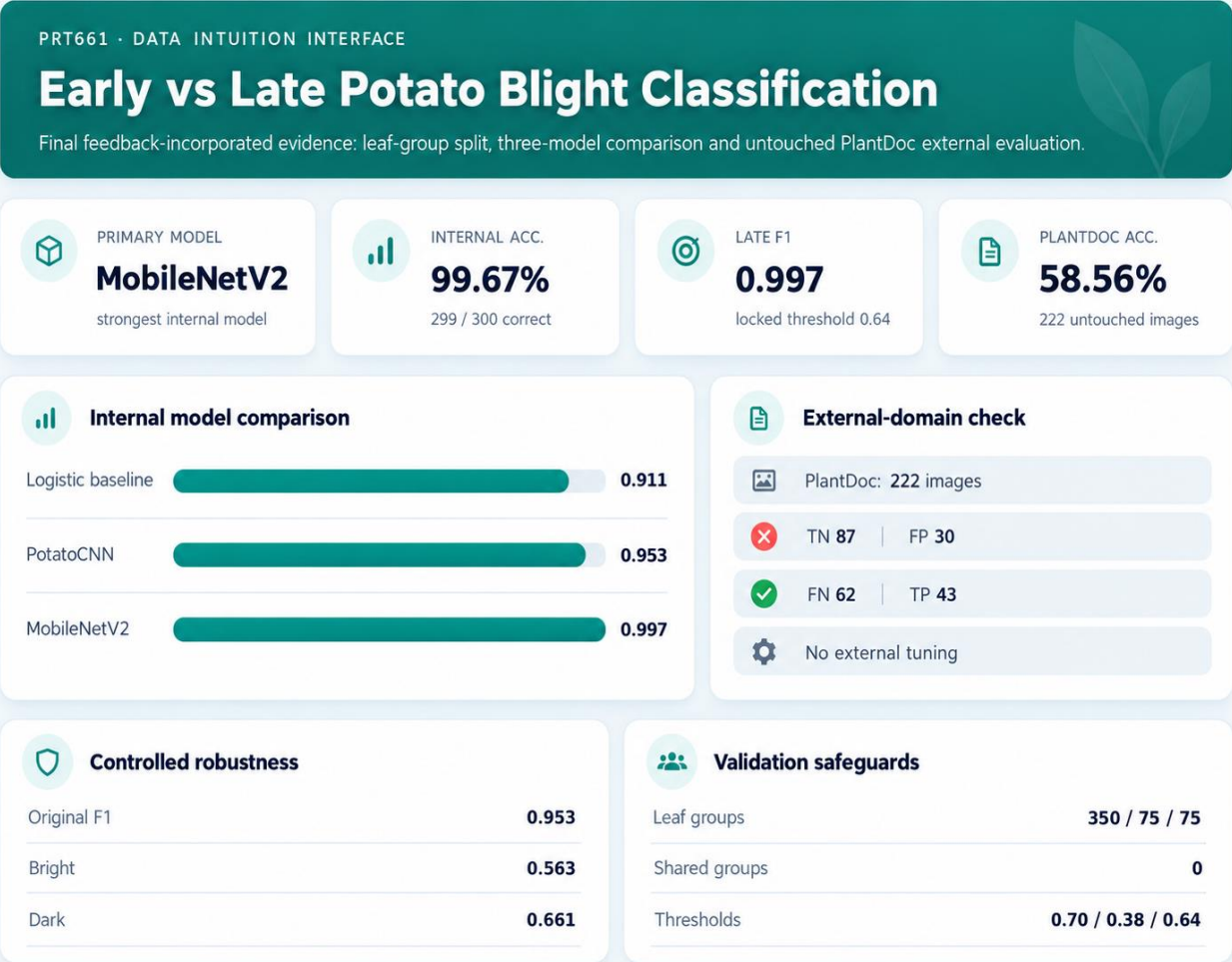


Ablation and efficiency

- Short ablations reached F1 0.9935–0.9967; the spread is about one test image.
- No strong augmentation/dropout architecture-winner claim is justified from that difference.
- PotatoCNN size: 0.360 MB · measured latency 0.819 ms/image · throughput 1,222 images/s.
- Efficiency supports prototyping; representative target-device benchmarking is still required.



Feedback-integrated results explorer

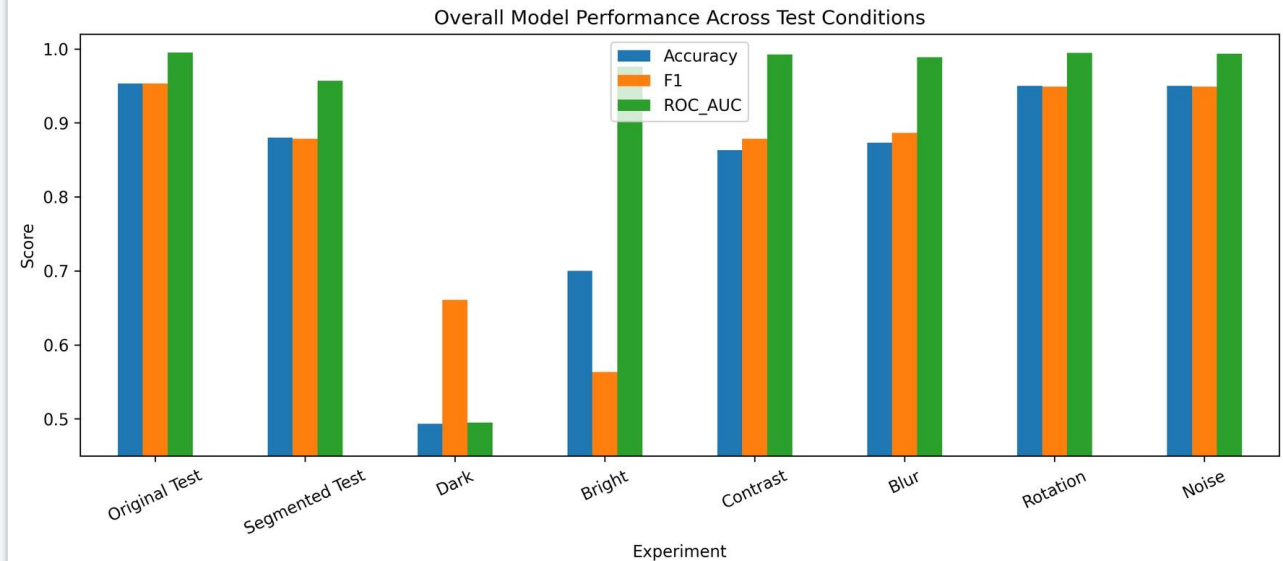


LIVE DEMO

- 1 Compare models
- 2 Inspect PlantDoc
- 3 Review locked threshold
- 4 Review robustness

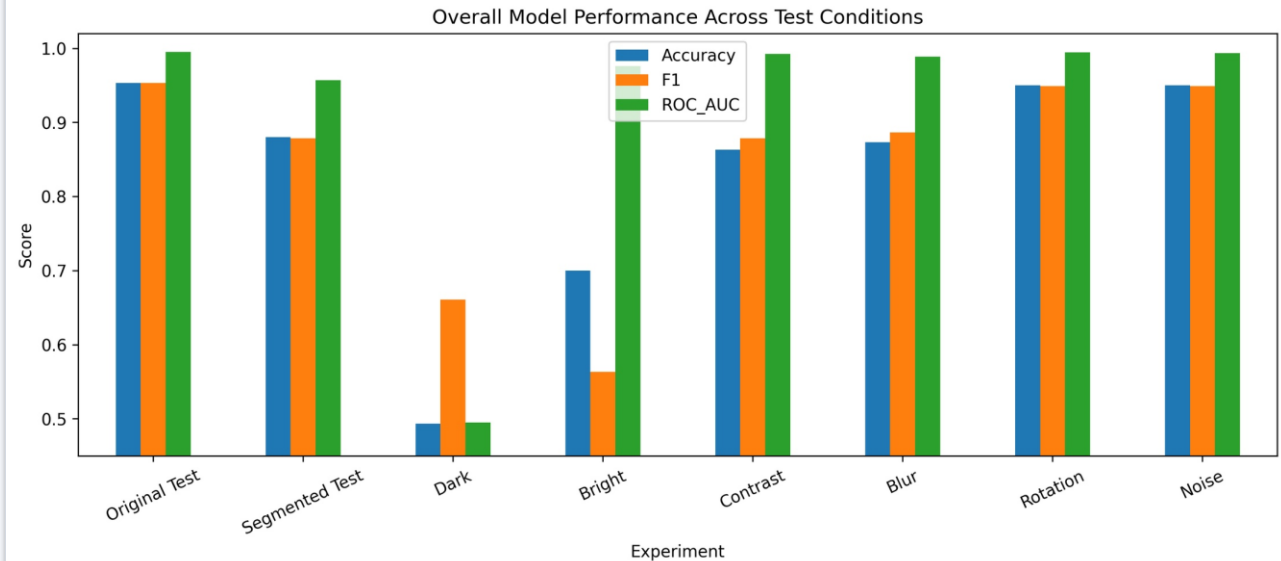
Ethics, privacy and security

- Two known disease labels do not equal field diagnosis or differential diagnosis.
- No pesticide, treatment or pathogen-confirmation output is produced.
- PlantDoc demonstrates external-domain weakness; field claims must remain explicitly limited.
- Uploads, model version, preprocessing and threshold require validation, logging and rollback.



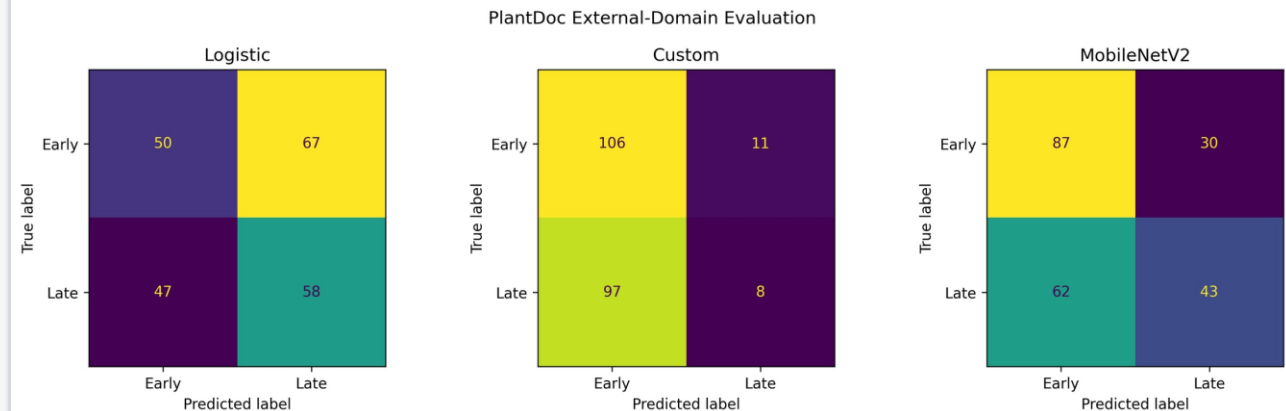
Implementation status and evidence gaps

- Complete: provenance, 500-leaf group split, logistic baseline, PotatoCNN and MobileNetV2.
- Complete: locked internal comparison, McNemar tests, XAI, calibration, segmentation, robustness, ablation and efficiency.
- Complete: untouched PlantDoc external evaluation plus versioned evidence/interface artefacts.
- Remaining submission items are team-specific links, contribution records and authentic project-management evidence.



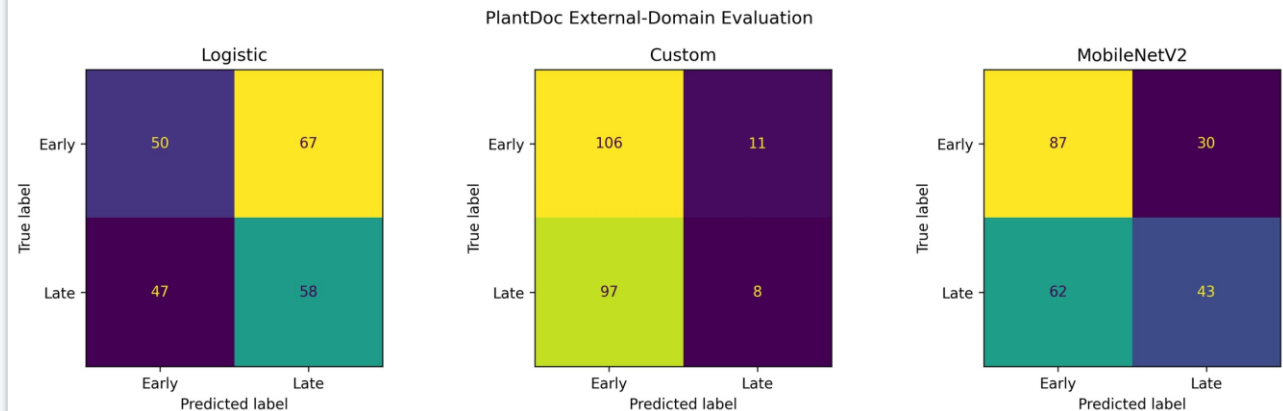
Prioritised next steps

- Collect prospective expert-labelled field images grouped by leaf, farm, device and site.
- Add Healthy, Other/Unknown and input-quality rejection or abstention.
- Keep preprocessing and thresholds versioned together; any representation change requires fresh validation.
- Calibrate prospectively, declare cost-based operating points and evaluate human review before field use.



Future work

- Finalise authentic GitHub/Jira evidence and individual contribution records.
- Reproduce the full pipeline in a clean environment with versioned dependencies.
- Validate field/generalisation using grouped images by leaf, farm, device and site.
- Prepare Assessment 4 demo/report with human review, rejection and calibration limits.



Conclusion

TEST ACCURACY

99.67%

group-isolated internal set

LATE RECALL

99.32%

147 / 148

PLANTDOC ACC.

58.56%

222 untouched images

PLANTDOC F1

0.4831

ROC-AUC 0.6210

TAKEAWAY

Excellent internal separation · weak PlantDoc transfer · field validation required before use

Selected references

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- ✓ Mohanty et al. (2016). Plant disease detection. *Frontiers in Plant Science*.
 - ✓ Hughes & Salathé (2015). PlantVillage repository announcement. *arXiv*.
 - ✓ Singh et al. (2020). PlantDoc: visual plant disease detection dataset. *CoDS-COMAD*.
 - ✓ Ferentinos (2018). Deep learning for plant disease. *Computers and Electronics in Agriculture*.
 - ✓ Sandler et al. (2018). MobileNetV2: inverted residuals and linear bottlenecks. *CVPR*.
 - ✓ Selvaraju et al. (2017). Grad-CAM. *ICCV*.
 - ✓ Guo et al. (2017). Neural-network calibration. *ICML*.
 - ✓ Hendrycks & Dietterich (2019). Corruption robustness. *ICLR*.